

DEKALB, IL, USA

WMO: 722075

Lat: 41.932N

Lon: 88.708W

Elev: 279

StdP: 98.02

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04871

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.9	-17.3	-24.6	0.4	-19.1	-21.8	0.6	-16.5	12.8	-2.6	11.7	-3.1	4.7	280	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	31.8	23.4	30.2	22.9	28.8	22.3	25.7	29.6	24.6	28.4	23.6	27.3	5.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	20.3	28.4	23.2	18.6	26.9	22.4	17.7	26.2	80.7	29.7	75.7	28.4	71.9	27.4	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	10.4	9.0	-23.7	33.5	3.7	1.7	-26.3	34.8	-28.5	35.8	-30.5	36.8	-33.2	38.1		
(5)			-24.0	27.3	3.5	1.0	-26.6	28.1	-28.6	28.7	-30.6	29.2	-33.2	30.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-5.2	-4.2	2.8	9.1	15.3	20.7	22.3	21.2	17.9	11.0	3.8	-2.4		
	DBStd	11.03	6.64	6.50	6.28	4.98	4.91	3.38	3.16	2.85	4.17	5.04	5.27	5.89		
	HDD10.0	1823	470	399	239	75	9	0	0	0	1	48	195	386		
	HDD18.3	3640	728	632	484	278	120	15	4	8	57	233	436	644		
	CDD10.0	1616	0	1	15	49	174	319	382	346	239	80	9	1		
	CDD18.3	391	0	0	2	2	27	85	128	96	45	7	0	0		
	CDH23.3	3256	0	0	11	28	262	709	1126	697	365	58	0	0		
	CDH26.7	900	0	0	1	1	68	211	348	164	97	10	0	0		
(14) Wind	WSAvg	4.3	4.9	4.9	4.9	5.3	4.6	3.9	3.2	2.9	3.4	4.2	4.7	4.8		
(15) Precipitation	PrecAvg	878	47	40	55	80	113	102	94	99	72	71	64	41		
	PrecMax	1102	83	131	105	137	204	239	183	256	154	168	138	77		
	PrecMin	551	13	12	22	38	21	26	26	26	15	11	5	9		
	PrecStd	128	21	27	28	33	55	57	44	49	42	46	39	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	15.2	25.1	26.4	31.5	32.6	34.9	32.4	31.9	27.8	19.9	14.7		
		MCWB	10.2	11.1	17.4	16.1	20.2	23.0	25.7	23.6	23.6	20.8	13.8	13.4		
	2%	DB	7.8	9.7	18.8	23.4	28.7	31.1	31.9	30.1	29.9	24.9	16.4	10.2		
		MCWB	6.6	7.8	13.6	15.4	19.6	22.5	25.1	23.6	22.2	17.6	12.3	8.7		
	5%	DB	5.1	6.0	15.2	21.1	26.5	29.2	30.0	28.4	27.3	22.1	14.0	7.5		
		MCWB	3.6	4.0	11.2	13.3	18.2	21.6	23.8	22.6	20.3	16.6	11.1	6.2		
	10%	DB	2.9	3.3	12.0	18.0	24.0	27.4	28.4	27.1	25.4	19.4	11.9	5.0		
		MCWB	1.9	2.0	8.8	12.4	17.8	20.7	22.9	21.7	19.4	15.7	9.1	3.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	11.9	17.5	18.6	23.2	26.0	27.4	26.3	25.2	22.4	15.9	13.6		
		MCDB	11.8	13.3	23.8	23.0	28.3	29.5	31.6	29.7	29.4	26.5	18.0	14.7		
	2%	WB	6.8	7.8	14.8	16.5	21.4	24.4	26.2	24.8	23.6	19.9	13.5	8.7		
		MCDB	7.8	9.4	18.5	21.1	26.1	28.5	30.3	28.5	27.6	22.5	15.6	9.7		
	5%	WB	3.8	4.5	12.4	15.0	20.0	23.4	25.1	23.7	22.2	17.9	11.3	6.1		
		MCDB	5.1	5.9	15.1	19.4	24.6	27.5	28.9	27.2	25.7	20.5	13.5	7.2		
	10%	WB	2.0	2.2	8.7	13.0	18.7	22.2	23.9	22.6	20.8	15.7	9.2	3.9		
		MCDB	3.0	3.4	11.5	17.9	22.6	26.0	27.4	25.8	23.9	18.6	11.6	5.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	7.6	8.9	11.2	11.1	10.5	10.3	10.1	11.1	10.2	8.4	6.9		
		MCDBR	8.5	10.2	13.1	15.5	13.7	12.3	11.4	11.4	12.7	13.3	11.9	9.3		
	5% WB	MCWBR	7.3	8.1	9.1	8.9	7.2	6.3	6.0	6.3	6.5	7.6	8.4	8.2		
		MCWBR	8.2	10.1	12.0	13.4	11.4	10.7	10.5	9.8	10.6	11.1	10.8	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.285	0.303	0.334	0.387	0.413	0.424	0.435	0.420	0.393	0.341	0.300	0.282		
		taud	2.416	2.379	2.403	2.264	2.243	2.254	2.215	2.250	2.303	2.450	2.510	2.450		
	Ebn at Noon		868	905	914	881	863	852	838	839	837	849	846	840		
		Edh at Noon	78	95	105	130	136	135	139	130	115	88	71	69		
(44) All-Sky Solar Radiation	RadAvg		1.67	2.60	3.63	4.65	5.35	6.08	6.08	5.34	4.46	2.92	1.89	1.39		
	RadStd		0.19	0.33	0.31	0.38	0.40	0.44	0.34	0.32	0.38	0.34	0.17	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page